

Module 23.14: Evaluation Metrics for Regression (MAE, MSE, RMSE, R^2 , and Adjusted R^2)

1. Why Classification Accuracy Fails for Regression

In classification problems, target outputs are discrete labels ($y \in \{0, 1\}$), permitting simple metric ratios such as $\frac{\text{Correct}}{\text{Total}}$. In regression, predictions $\hat{y}$ are continuous real numbers ($\hat{y} \in \mathbb{R}$). The probability of an exact floating-point match $P(\hat{y} = y) = 0$. Consequently, regression evaluation relies on **distance-based error norms** and **variance-explained ratios**.

2. Distance-Based Error Metrics

1. Mean Absolute Error (MAE)

MAE measures the average absolute magnitude of the prediction errors:

$$\text{MAE} = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i|$$

- **Units:** Exact same units as the target variable y .
- **Robustness:** Robust to outliers because errors scale linearly ($|e_i|$). It treats all deviations proportionally.

2. Mean Squared Error (MSE)

MSE computes the average of the squared prediction deviations:

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

- **Units:** Squared units of the target (y^2 , e.g., $\text{\2).

- **Outlier Sensitivity:** Highly sensitive to outliers. A single severe error drastically inflates MSE due to the quadratic exponent.

3. Root Mean Squared Error (RMSE)

RMSE is the square root of the Mean Squared Error:

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}$$

- **Units:** Restores the original target unit y while preserving the quadratic penalty on large outliers.
- **Mathematical Property:** $\text{RMSE} \geq \text{MAE}$ always. The difference $\text{RMSE} - \text{MAE}$ reflects the variance in error magnitudes.

3. Relative Goodness-of-Fit: R^2 Score

The **Coefficient of Determination** (R^2) quantifies the proportion of target variance explained by the model compared to a baseline naive model that always predicts the sample mean $\bar{y}$.

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y})^2}$$

Where:

- SS_{res} (**Residual Sum of Squares**): Unexplained variance in predictions.
- SS_{tot} (**Total Sum of Squares**): Total empirical variance in ground truth y .

Value of R^2	Interpretation
$R^2 = 1.0$	Perfect model (zero residual errors; all points lie precisely
$R^2 = 0.0$	Model performs identically to predicting the sample mean $\bar{y}$
$R^2 < 0.0$	Model performs WORSE than simply predicting the mean!

4. The Pitfall of R^2 and the Adjusted R^2 Solution

The Fundamental Flaw of R^2 :

As additional feature variables are added to a regression model ($p \rightarrow p + 1$), SS_{res} can **never increase**; it either decreases or stays constant. Even if a completely random, useless noise feature is added, R^2 will artificially increase or remain unchanged. It never penalizes model bloat!

The Adjusted R^2 Metric:

Adjusted R^2 introduces a degrees-of-freedom penalty for every additional predictor feature p :

$$R^2_{\text{adj}} = 1 - \left[\frac{(1 - R^2)(m - 1)}{m - p - 1} \right]$$

Where:

- m : Total sample size.
- p : Total number of independent predictors/features.

Behavioral Properties:

1. If a newly added feature meaningfully reduces SS_{res} beyond chance, R^2_{adj} **increases**.
 2. If a newly added feature offers negligible predictive value, the penalty term $\frac{m-1}{m-p-1}$ dominates, and R^2_{adj} **decreases**.
 3. $R^2_{\text{adj}} \leq R^2$ always holds.
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5. Comparative Evaluation Summary

Metric	Formula	Units	Sensitivity to Outliers	Primary Use Case
MAE	$\frac{1}{m} \sum y_i - \hat{y}_i $	Same as y	Low (Robust)	Business metrics requiring intuitive linear interpretation
MSE	$\frac{1}{m} \sum (y_i - \hat{y}_i)^2$	Squared y^2	High	Loss function during model training optimization
RMSE	$\sqrt{\text{MSE}}$	Same as y	High	Diagnostic evaluation when large errors are dangerous
R^2	$1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$	Unitless $\in (-\infty, 1]$	Medium	Evaluating explanatory variance of univariate models
R^2_{adj}	$1 - \frac{(1-R^2)(m-1)}{m-p-1}$	Unitless $\in (-\infty, 1]$	Medium	Comparing multiple regression models with different feature counts